

Feasibility Memo: AI-Assisted Emergency Department Triage Dataset

To: Dr. De Fretias and Emergency Department Board

From: Skye Thomas

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Subject: Feasibility Assessment of Emergency Department Triage Dataset for Baseline AI Modelling

1. One-Sentence Verdict

Verdict: The emergency department dataset is suitable for developing a baseline triage prediction model; however, several data quality and clinical governance considerations should be addressed before any clinical implementation.

2. Dataset Summary

The dataset provided for this project contains **55,121 patient encounters** and **226 variables** extracted from an emergency department electronic health record (EHR) system. The variables encompass several clinically relevant domains including:

- Patient demographics
- Triage acuity levels
- Vital signs
- Clinical assessments
- Chief complaints
- Admission outcomes
- Administrative information

Initial exploratory data analysis indicates that the dataset is well-structured and highly complete. No missing values were identified across any of the 226 variables. This level of completeness is unusual for operational emergency department datasets and suggests that the dataset has undergone substantial preprocessing or curation prior to release.

The dataset appears sufficiently large to support exploratory modelling and baseline machine learning experimentation. Furthermore, the inclusion of multiple physiological and administrative variables provides a strong foundation for predicting patient triage outcomes.

3. Top Three Data Quality Concerns

Concern 1: Absence of Missing Data

No missing values were identified within the dataset. While advantageous for machine learning applications, complete absence of missingness is uncommon in real-world emergency department environments. This raises questions regarding prior data cleaning, imputation strategies, or exclusion criteria that have not been documented.

Potential impact: Unknown preprocessing procedures may affect model generalizability to operational clinical settings.

Concern 2: Potential Single-Site Bias

The dataset appears to represent patient encounters from a single healthcare institution. Patient demographics, disease prevalence, and triage practices can vary substantially across hospitals and geographic regions.

Potential impact: Models trained on this dataset may perform poorly when deployed in other healthcare environments.

Concern 3: High-Dimensional Feature Space

The dataset contains 226 variables, many of which represent chief complaints and clinical observations. Some variables may be redundant, highly correlated, or clinically irrelevant to triage prediction.

Potential impact: Excessive feature dimensionality may increase model complexity and reduce interpretability.

4. Top Three Reasons to Proceed

Reason 1: Large Patient Population

The dataset contains over 55,000 emergency department encounters, providing sufficient observations to train and validate baseline predictive models while reducing the likelihood of overfitting.

Reason 2: Clinically Relevant Variables

The dataset contains multiple physiological measurements commonly used in emergency triage, including vital signs and presenting complaints. These variables align closely with established clinical triage workflows.

Reason 3: Excellent Data Completeness

The absence of missing values eliminates the need for extensive imputation strategies during baseline model development and allows the project to focus on feature engineering and predictive performance.

5. Preliminary Feature Shortlist

Based on clinical intuition and established emergency medicine practice, the following variables are likely to be among the strongest predictors of triage level:

Rank	Feature	Clinical Justification
1	Respiratory Rate	Elevated respiratory rate is a strong indicator of physiological distress.
2	Heart Rate	Tachycardia frequently indicates acute illness or instability.
3	Oxygen Saturation	Low oxygen saturation reflects respiratory compromise requiring urgent assessment.
4	Systolic Blood Pressure	Blood pressure abnormalities indicate cardiovascular instability.
5	Temperature	Fever and hypothermia are markers of systemic illness.

6	Age	Patient age significantly influences clinical risk stratification.
7	Pain Score	Severe pain often influences triage prioritization.
8	Chief Complaint	Presenting complaint provides essential clinical context.
9	Mental Status/Consciousness	Altered consciousness is associated with high-acuity presentations.
10	Mode of Arrival	Ambulance arrival often correlates with increased illness severity.

6. Planned Data Quality Visualizations

The exploratory analysis includes the following visualizations:

- Distribution of triage levels
- Distribution of patient race and ethnicity
- Frequency analysis of chief complaints
- Correlation heatmap of selected clinical variables
- Vital sign distribution plots
- Outlier analysis for physiological measurements

7. Caveats and Limitations

Several limitations should be considered before proceeding to clinical deployment:

- The dataset appears to represent a single clinical site.
- Prior preprocessing methods have not been fully documented.
- Real-world emergency department data typically contain missing information that is absent in this dataset.
- Clinical validation and fairness assessment will be required before implementation.
- Model performance alone should not determine clinical adoption decisions.

8. Recommendation

Based on the current exploratory analysis, the dataset is considered suitable for developing a baseline emergency department triage prediction model. The large sample size, clinically relevant variables, and high level of data completeness support proceeding to model development. However, additional investigation into data provenance, preprocessing history, and external validity should be undertaken before considering operational deployment.